

List of requirements for healthcare

- Disease with most number of claims

max_claims_disease : X subscriber_age_group : X hospital_most_patie... : X subgroup_subscribe... : X claims_rejection : X claims_by_city : X government_v

```
1 -- Disease with Maximum Claims (Approved only)
2 SELECT
3     disease_name,
4     COUNT(*) as total_claims,
5     SUM(claim_amount) as total_claim_amount,
6     AVG(claim_amount) as avg_claim_amount
7 FROM healthcare_db.claims
8 WHERE claim_status = 'Approved'
9 GROUP BY disease_name
10 ORDER BY total_claims DESC
11 LIMIT 10;
12
13 -- 1. Which disease has a maximum number of claims?
14 SELECT
15     disease_name,
```

SQL Ln 1, Col 3

Run Explain Explain Explain Cancel Clear Create

Query results Query status

Completed Time in queue: 70 ms Run time:

Results (10)

Search rows

#	disease_name	total_claims	total_claim_amount	avg_claim_amount
1	Galactosemia	2	198502	99251.0
2	Cholera	2	238236	119118.0

- Subscribers having age less than 30 and they subscribe any subgroup & group with maximum subgroup

max_claims_disease : X subscriber_age_group : X hospital_most_patie... : X subgroup_subscribe... : X claims_rejection : X claims_by_city

```
1 -- ===== PATIENT & SUBSCRIBER ANALYSIS =====
2
3 -- 11. Age Group Analysis
4 SELECT
5     CASE
6         WHEN age < 18 THEN '0-17 (Minor)'
7         WHEN age BETWEEN 18 AND 30 THEN '18-30 (Young Adult)'
8         WHEN age BETWEEN 31 AND 45 THEN '31-45 (Adult)'
9         WHEN age BETWEEN 46 AND 60 THEN '46-60 (Middle Age)'
10        ELSE '60+ (Senior)'
11    END as age_group,
12    COUNT(*) as patient_count,
13    AVG(age) as avg_age
14 FROM healthcare_db.patient_records
15 WHERE age IS NOT NULL
SQL Ln 1, Col 1
```

Run again Explain Explain Explain Cancel Clear Create

Query results Query status

Completed Time in que

Results (4)

Search rows

#	age_group	patient_count	avg_age
1	60+ (Senior)	46	84.41304347826087
2	46-60 (Middle Age)	9	55.22222222222222
3	31-45 (Adult)	14	38.42857142857143

- Find out hospital which serve most number of patients

SQL Editor interface showing a query to find the hospital serving the most number of patients.

```

1 --hospital serving the most number of patients
2
3 SELECT
4   h.hospital_id,
5   h.hospital_name,
6   COUNT(p.patient_id) AS patient_count
7 FROM healthcare_db.hospital h
8 JOIN healthcare_db.patient_records p
9   ON h.hospital_id = p.hospital_id
10 GROUP BY h.hospital_id, h.hospital_name
11 ORDER BY patient_count DESC
12 LIMIT 5;
13

```

SQL Ln 12, Col 8

Buttons: Run again, Explain ⁱ, Cancel, Clear, Create ▼

Query results | Query status

Completed Time in queue: 91 ms Run

Results (5)

Search rows

#	hospital_id	hospital_name	patient_count
1	H1017	Manipal Hospitals	9
2	H1019	Apollo Hospitals - Bannerghatta Road	8
3	H1001	Medanta The Medicity	7

- Find out which subgroups subscribe most number of times

SQL Editor interface showing a query to find which subgroups subscribe most number of times.

```

1 -- 5. Find out which subgroups subscribe most number of times
2 SELECT
3   sg.subgrp_id,
4   sg.subgrp_name,
5   sg.monthly_premium,
6   COUNT(s.sub_id) as subscriber_count
7 FROM healthcare_db.subgroup sg
8 LEFT JOIN healthcare_db.subscriber s ON sg.subgrp_id = s.subgrp_id
9 GROUP BY sg.subgrp_id, sg.subgrp_name, sg.monthly_premium
10 ORDER BY subscriber_count DESC;

```

SQL Ln 7, Col 28

Buttons: Run again, Explain ⁱ, Cancel, Clear, Create ▼

Query results | Query status

Completed Time in queue: 55 ms Run

Results (10)

Search rows

#	subgrp_id	subgrp_name	monthly_premium	subscriber_count
1	S104	Therapy	1500	13
2	S109	Hereditary	2000	11
3	S101	Deficiency Diseases	3000	11

- Find out total number of claims which were rejected

SQL Editor interface showing a query to find the total number of claims which were rejected.

```

1 -- 6. Find out total number of claims which were rejected
2 SELECT
3   claim_status,
4   COUNT(*) as total_rejected_claims,
5   SUM(claim_amount) as total_rejected_amount,
6   AVG(claim_amount) as avg_rejected_amount
7 FROM healthcare_db.claims
8 WHERE claim_status = 'Rejected'
9 GROUP BY claim_status;

```

SQL Ln 7, Col 26

Buttons: Run again, Explain ⁱ, Cancel, Clear, Create ▼

Query results | Query status

Completed Time in queue: 93 ms Run time

Results (1)

Search rows

#	claim_status	total_rejected_claims	total_rejected_amount	avg_rejected_amount
1	Rejected	18	1708003	94889.05555555556

- From where most claims are coming (city)

SQL Editor interface showing a query to find the city with the most claims.

```

1 -- 7. From where most claims are coming (city)
2 SELECT
3   p.city,
4   COUNT(c.claim_id) as total_claims,
5   SUM(c.claim_amount) as total_claim_amount,
6   AVG(c.claim_amount) as avg_claim_amount
7 FROM healthcare_db.claims c
8 JOIN healthcare_db.patient_records p ON c.patient_id = p.patient_id
9 GROUP BY p.city
10 ORDER BY total_claims DESC
11 LIMIT 10;

```

Query results (10):

#	city	total_claims	total_claim_amount	avg_claim_amount
1	Amravati	2	146491	73245.5
2	Mysore	2	71510	35755.0
3	Kamarhati	2	190252	95126.0

- Which groups of policies subscriber subscribe mostly Government or private

SQL Editor interface showing a query to analyze policy groups by subscriber type.

```

1 -- 15. Government vs Private Group Analysis
2 SELECT
3   g.grp_type,
4   COUNT(DISTINCT g.grp_id) as total_groups,
5   SUM(g.premium_written) as total_premium,
6   AVG(g.premium_written) as avg_premium_per_group,
7   COUNT(DISTINCT gs.subgrp_id) as total_subgroups
8 FROM healthcare_db."group" g
9 LEFT JOIN healthcare_db.grpsubgrp gs ON g.grp_id = gs.grp_id
10 GROUP BY g.grp_type;

```

Query results (2):

#	grp_type	total_groups	total_premium	avg_premium_per_group	total_subgroups
1	Government	7	434000	62000.0	3
2	Private	51	3502000	6851.851851851854	10

- Average monthly premium subscriber pay to insurance company.

SQL Editor interface showing a query to calculate average monthly premium.

```

1 -- 9. Average monthly premium subscriber pay to insurance company
2 SELECT
3   ROUND(AVG(sg.monthly_premium), 2) as avg_monthly_premium,
4   MIN(sg.monthly_premium) as min_premium,
5   MAX(sg.monthly_premium) as max_premium,
6   COUNT(DISTINCT s.sub_id) as total_subscribers
7 FROM healthcare_db.subscriber s
8 JOIN healthcare_db.subgroup sg ON s.subgrp_id = sg.subgrp_id;

```

Query results (1):

#	avg_monthly_premium	min_premium	max_premium	total_subscribers
1	1867.35	1000	3200	98

- Find out Which group is most profitable

```

1  -- 10. Find out Which group is most profitable
2  -- (Premium collected - Claims paid)
3  WITH group_revenue AS (
4      SELECT g.grp_id,
5             g.grp_name,
6             g.grp_type,
7             SUM(
8                 sg.monthly_premium * date_diff(
9                     'month',
10                    s.eff_date,
11                    COALESCE(s.term_date, CURRENT_DATE)
12                )
13            ) as total_premium_collected
14  FROM healthcare_db."group" g
15  JOIN healthcare_db.grpsubgrp gs ON g.grp_id = gs.grp_id
16  JOIN healthcare_db.subscriber s ON gs.subgrp_id = s.subgrp_id
17  JOIN healthcare_db.subgroup sg ON s.subgrp_id = sg.subgrp_id
18  GROUP BY g.grp_id,
19             g.grp_name,
20             g.grp_type
21  ),
22  group_claims AS (
23      SELECT g.grp_id,
24             SUM(c.claim_amount) as total_claims_paid
25  FROM healthcare_db."group" g
26  JOIN healthcare_db.grpsubgrp gs ON g.grp_id = gs.grp_id
27  JOIN healthcare_db.subscriber s ON gs.subgrp_id = s.subgrp_id
28  JOIN healthcare_db.claims c ON s.sub_id = c.sub_id
29  WHERE c.claim_status = 'Approved'
30  GROUP BY g.grp_id
31  )
32  SELECT r.grp_id,
33         r.grp_name,
34         r.grp_type,
35         r.total_premium_collected,
36         COALESCE(c.total_claims_paid, 0) as total_claims_paid,
37         (
38             r.total_premium_collected - COALESCE(c.total_claims_paid, 0)
39         ) as profit
40  FROM group_revenue r
41  LEFT JOIN group_claims c ON r.grp_id = c.grp_id
42  ORDER BY profit DESC
43  LIMIT 5;

```

SQL Ln 15, Col 58

Run Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query status

Completed Time in queue: 98 ms Run time: 953 ms Data scanned: 4.39 KB

Results (5)

Search rows

#	grp_id	grp_name	grp_type	total_premium_collected	total_claims_paid	profit
1	GRP101	Life Insurance Corporation of India	Government	2436000	156560	2279440
2	GRP105	Kotak Mahindra Life Insurance Co. Ltd.	Private	2436000	156560	2279440
3	GRP141	Kotak Mahindra General Insurance	Private	1891200	0	1891200
4	GRP131	Cholamandalam MS General Insurance	Private	1891200	0	1891200
5	GRP121	Pramerica Life Insurance Co. Ltd.	Private	1891200	0	1891200

- List all the patients below age of 18 who admit for cancer

```

1 -- 11. List all the patients below age of 18 who admit for cancer
2 SELECT
3     p.patient_id,
4     p.patient_name,
5     p.patient_gender,
6     p.age,
7     p.disease_name,
8     p.city,
9     h.hospital_name
10 FROM healthcare_db.patient_records p
11 JOIN healthcare_db.hospital h ON p.hospital_id = h.hospital_id
12 WHERE p.age < 18
13 AND LOWER(p.disease_name) LIKE '%cancer%'
14 ORDER BY p.age;

```

SQL Ln 12, Col 14

Run again Explain L² Cancel Clear Create

Query results Query status

Completed Time in queue: 60 ms Run time: 664 ms Data scanned: 0.66 KB

Results (0)

Search rows

#	patient_id	patient_name	patient_gender	age	disease_name	city	hospital_name
No results							

Run a query to view results

- List patients who have cashless insurance and have total charges greater than or equal for Rs. 50,000.

```

1 -- 12. List patients who have cashless insurance and have total charges greater than or equal to Rs. 50,000
2 SELECT
3     c.patient_id,
4     p.patient_name,
5     p.patient_gender,
6     p.age,
7     c.disease_name,
8     c.claim_amount,
9     c.claim_date,
10    s.elig_ind as cashless_eligible
11 FROM healthcare_db.claims c
12 JOIN healthcare_db.patient_records p ON c.patient_id = p.patient_id
13 JOIN healthcare_db.subscriber s ON c.sub_id = s.sub_id
14 WHERE s.elig_ind = 'Y' -- Eligible for cashless
15 AND c.claim_amount >= 50000
16 AND c.claim_status != 'Rejected'
17 ORDER BY c.claim_amount DESC
18 LIMIT 10;
19

```

SQL Ln 18, Col 9

Run again Explain L² Cancel Clear Create

Query results Query status

Completed Time in queue: 97 ms Run time: 823 ms Data scanned: 0.66 KB

Results (10)

Search rows

#	patient_id	patient_name	patient_gender	age	disease_name	claim_amount	claim_date	cashless_eligible
1	194166	Unknown	Male	79	Colorectal cancer	193801	1969-02-01	Y
2	189996	Ekant	Male	82	Measles	192381	1966-08-30	Y
3	108576	Chakrika	Male	61	Stroke	188727	1989-10-28	Y
4	132947	Saroj	Female	83	Anaemia	186502	1962-09-18	Y

- List female patients over the age of 40 that have undergone knee surgery in the past year

<
✕
claims_by_city : ✕
government_vs_priv... : ✕
avg_premiums : ✕
most_profitable_gro... : ✕
cancer_patients_u18 : ✕
✕

```

1 -- 13. List female patients over the age of 40 that have undergone knee surgery in the past year
2 SELECT
3     p.patient_id,
4     p.patient_name,
5     p.patient_gender,
6     p.age,
7     p.disease_name,
8     c.claim_date,
9     c.claim_amount,
10    h.hospital_name
11 FROM healthcare_db.patient_records p
12 JOIN healthcare_db.claims c ON p.patient_id = c.patient_id
13 JOIN healthcare_db.hospital h ON p.hospital_id = h.hospital_id
14 WHERE p.patient_gender = 'Female'
15     AND p.age > 40
16     AND (LOWER(p.disease_name) LIKE '%knee%'
17         OR LOWER(c.claim_type) LIKE '%surgery%'
18         OR LOWER(p.disease_name) LIKE '%arthritis%'
19         OR LOWER(p.disease_name) LIKE '%fracture%')

```

SQL Ln 15, Col 16

Run
Explain i?
Cancel
Clear
Create ▼

Query results
Query status

✔ Completed

Results (0)

Search rows

#	patient_id	patient_name	patient_gender	age	disease_name	claim_date
No results						